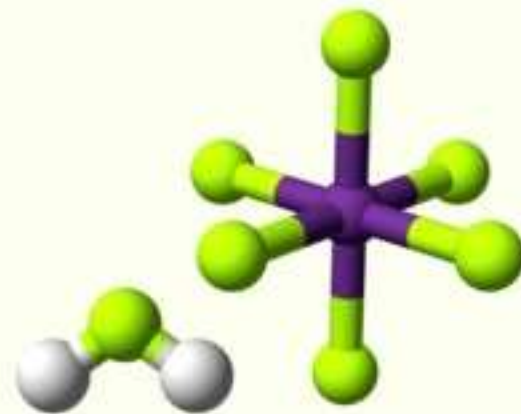
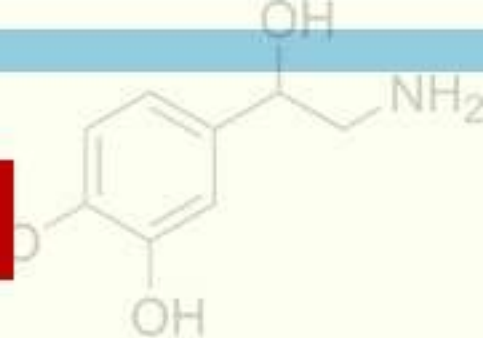


CHEMISTRY

ALL ABOUT IMPURE MATTER

MODULE - 3





IMPURE MATTER - CLASSIFICATION

IMPURE MATTER

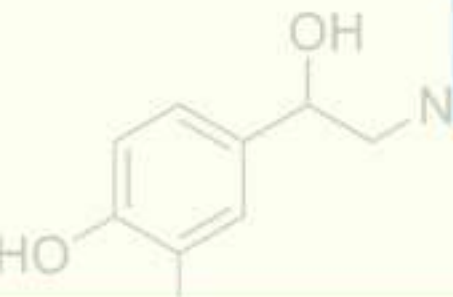
HOMOGENEOUS
MIXTURE

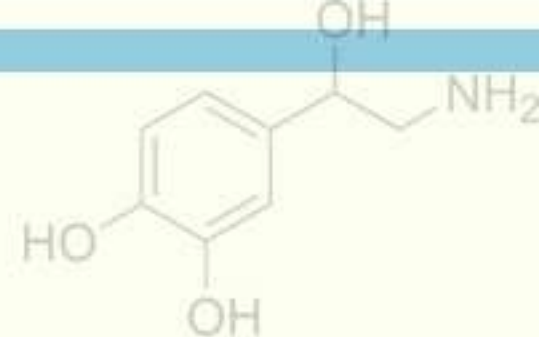
HETEROGENEOUS
MIXTURE

SOLUTION

SUSPENSION

COLLOID





COLLOIDS

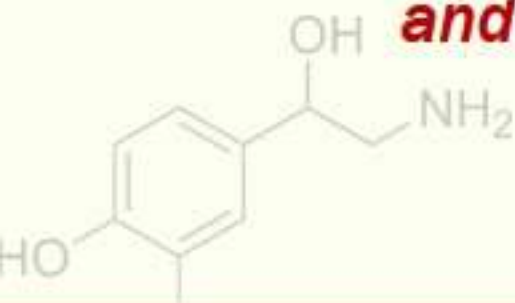
SOL

When a colloidal mixture appears like a fluid it is called a sol

GEL

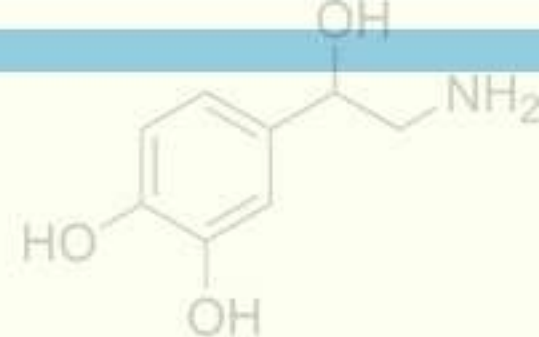
When a colloidal mixture appears like a solid it is called a gel

Colloids based on their appearance can be either Sol or Gel and can be classified on the basis of the physical states of the phase & medium



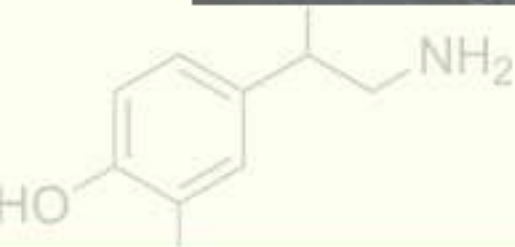


COLLOIDS - CLASSIFICATION



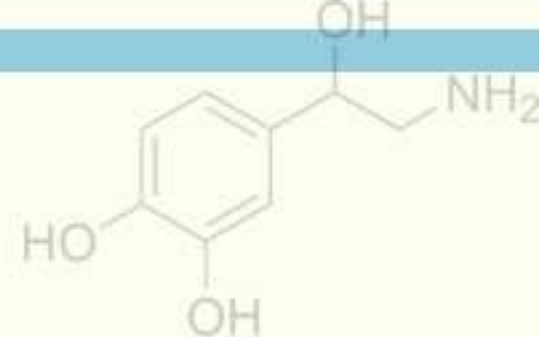
EXHAUST FUMES FROM AUTOMOBILES

- Smoke released by vehicles contain lot of soot and dust particles
- The solid particles remain uniformly suspended in the gaseous fumes
- The solid particles is the dispersed phase in the gas medium
- This type of colloid is called **AEROSOL**

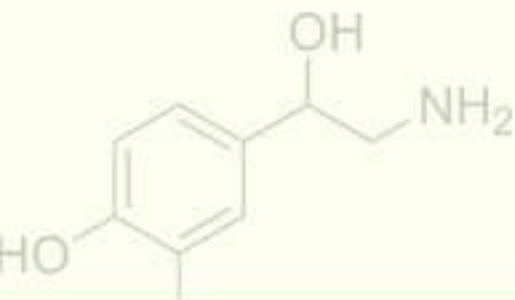




COLLOIDS - CLASSIFICATION

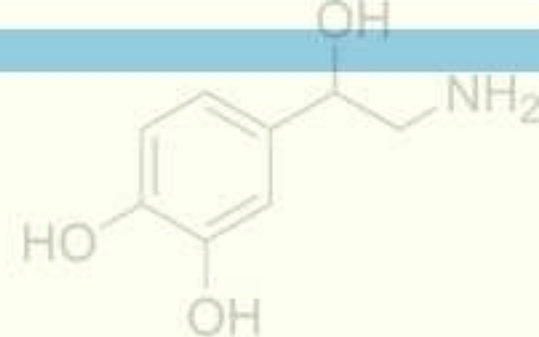


TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Solid Aerosol	Solid	Gas	- Smoke - Automobile exhaust



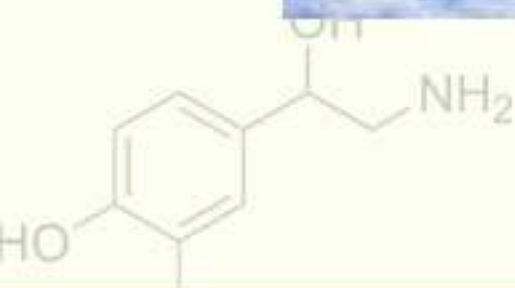


COLLOIDS - CLASSIFICATION



CLOUDS IN THE ATMOSPHERE

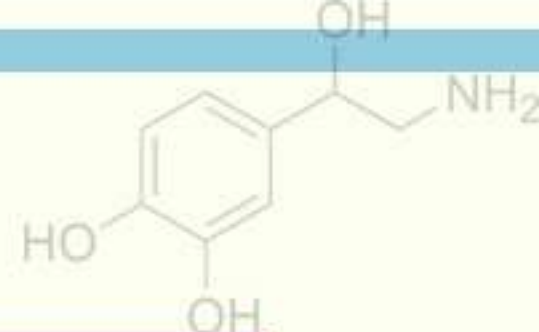
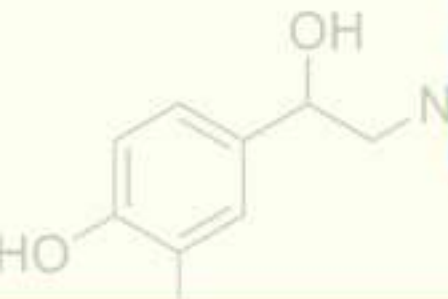
- Clouds are formed by the condensation of water vapor over dust particles in air
- Clouds are water molecules that is dispersed in the air
- The dispersed phase is liquid and the medium is gas i.e. air
- This is an example of a liquid **AEROSOL**





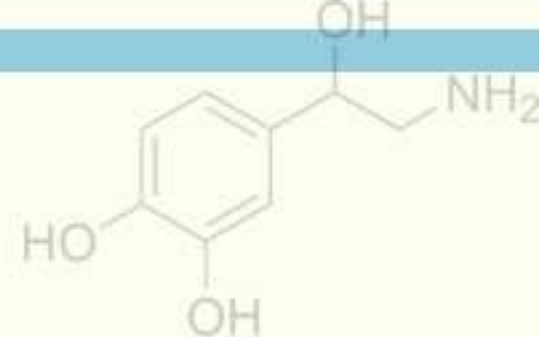
COLLOIDS - CLASSIFICATION

TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Liquid Aerosol	Liquid	Gas	■ Fog ■ Clouds ■ Mist



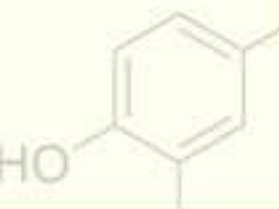


COLLOIDS - CLASSIFICATION



COLOURED GEMSTONES

- The gemstones are a valuable decorative piece in jewelry and astrology
- Gemstones are formed by one mineral mixing with other to give it a specific feature .
- A solid dispersed in another solid medium
- This is an example of a **SOLID SOL**





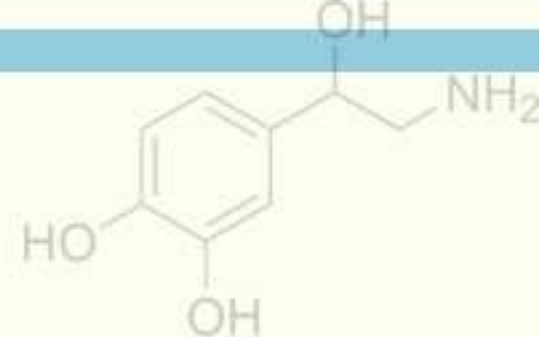
THE CLASSIFICATION OF COLLOIDS

TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Solid sol	Solid	Solid	- Colored Gemstone - Milky Glass





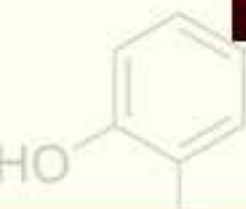
COLLOIDS - CLASSIFICATION

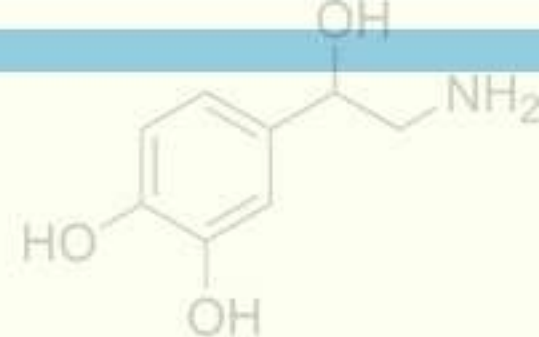


BLOOD – THE CONNECTIVE TISSUE



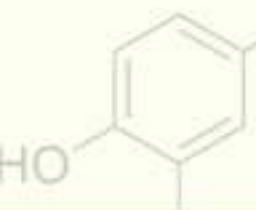
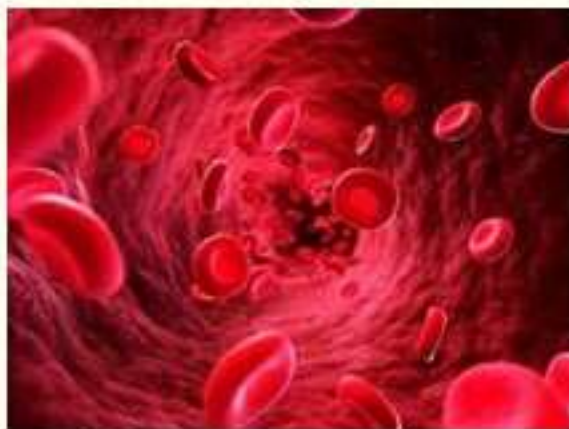
- The major circulatory fluid in our body
- Blood is made up of liquid blood plasma in which the blood cells are suspended.
- The blood cells are solid phase which is uniformly dispersed in liquid plasma
- This is an example of a liquid **SOL**





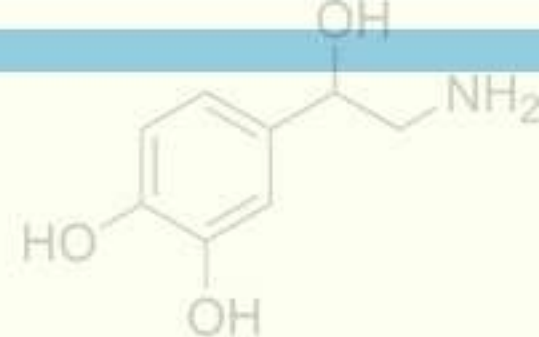
COLLOIDS - CLASSIFICATION

TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Liquid Sol	Solid	Liquid	▪ Blood ▪ Milk of magnesia ▪ Mud and dyes





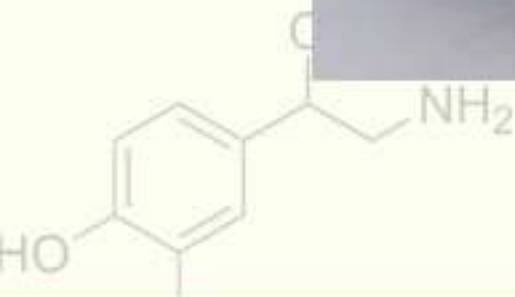
COLLOIDS - CLASSIFICATION

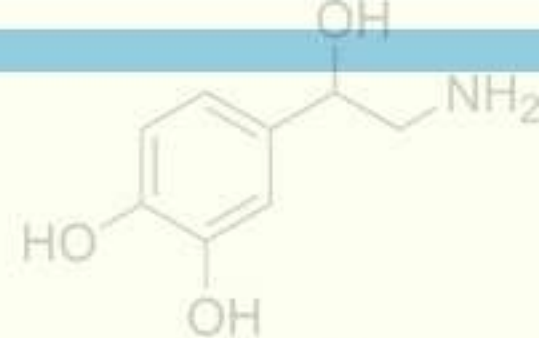


WHIPPED CREAM



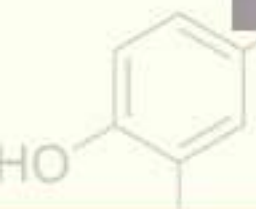
- Used in decoration of cakes and pastries
- It is made by inserting lots of air bubbles inside the creamy milk
- The gas particles are dispersed in liquid cream
- This is an example of a liquid **FOAM**





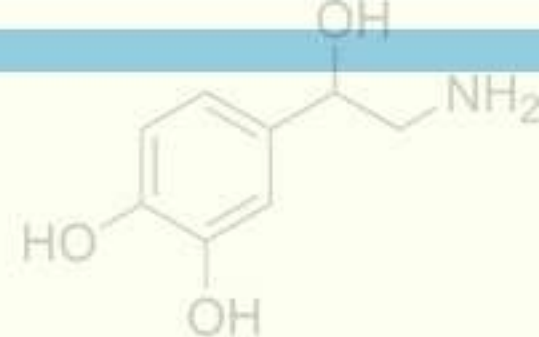
COLLOIDS - CLASSIFICATION

TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Liquid foam	Gas	Liquid	- Whipped cream - Shaving cream



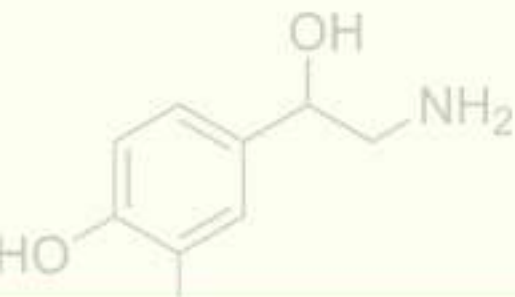


COLLOIDS - CLASSIFICATION



PUMICE STONE

- Pumice stone is very useful in scrubbing dead skin during bathing
- This stone has porous where air is trapped
- The gas is dispersed in the solid dispersing medium
- This is an example of a solid **FOAM**





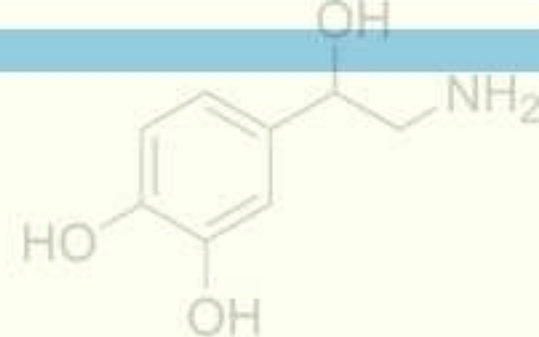
THE CLASSIFICATION OF COLLOIDS

TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Solid Foam	Gas	Solid	▪ Rubber ▪ Sponge ▪ Pumice ▪ Polystyrene





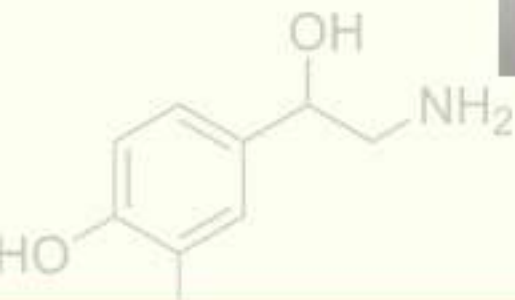
COLLOIDS - CLASSIFICATION



MILK – NATURE'S PERFECT FOOD



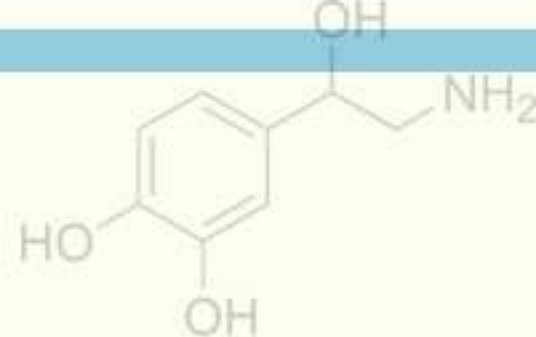
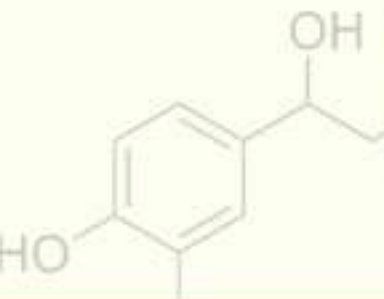
- Milk is a daily part of our diet
- Milk is made up of liquid oils and fats which is uniformly dispersed in water
- Thus the **immiscible liquid** oil is the dispersed phase in dispersing medium water
- This is an example of a **EMULSION**





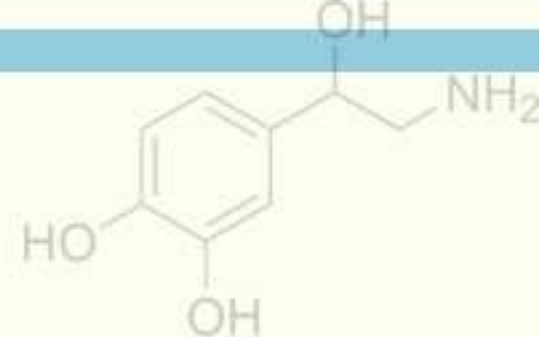
COLLOIDS - CLASSIFICATION

TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Emulsion	Liquid	Liquid	■ Mayonnaise ■ Milk ■ Face cream





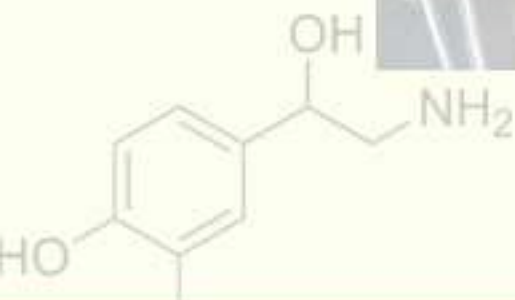
COLLOIDS - CLASSIFICATION



JELLY – FOR THE SWEET TOOTH

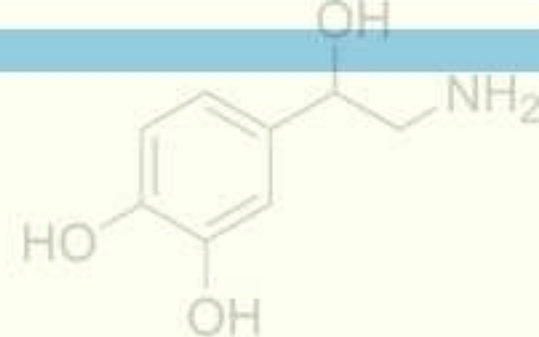


- A very tasty sweet items often loved by children
- Jellies are made when liquid syrup are mixed with gelatin to give it a shape
- Liquid particles are dispersed in solid medium
- This is an example of a **GEL**

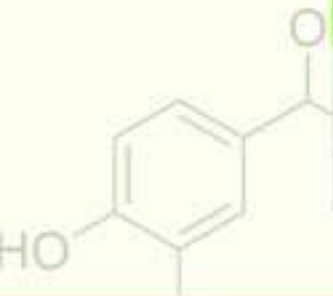




COLLOIDS - CLASSIFICATION



TYPE	DISPERSED PHASE	DISPERSING MEDIUM	EXAMPLES
Gel	Liquid	Solid	▪ Jelly ▪ Cheese ▪ Butter ▪ Toothpaste





COLLOIDS - CLASSIFICATION

BASED ON THE PHYSICAL STATES OF THE
DISPERSING MEDIUM

DISPERSING MEDIUM

GAS

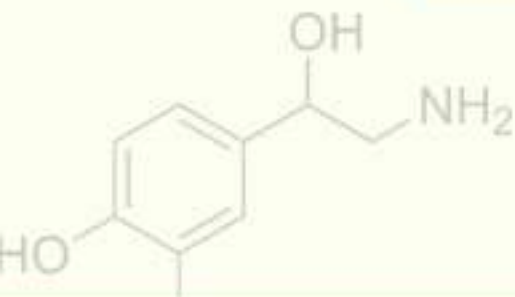
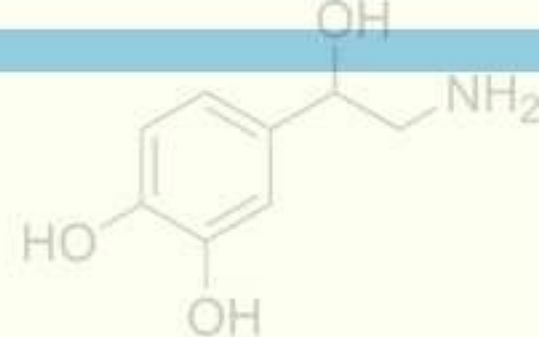
Aerosol

LIQUID

- Liquid Foam
- Emulsion
- Liquid Sol

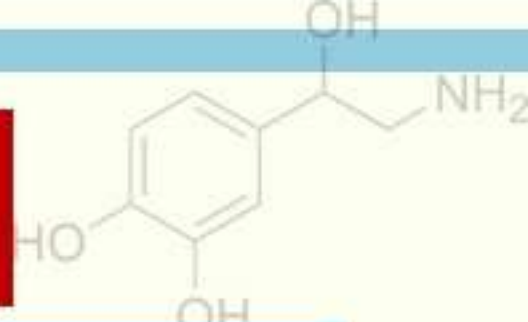
SOLID

- Solid Foam
- Gel
- Solid Sol





COLLOIDS BASED ON THE INTERACTION BETWEEN THE PHASES & MORE ...



1 HYDROPHILIC

Hydrophilic colloid-
strong attraction
between the
dispersed phase and
the medium
e.g. milk



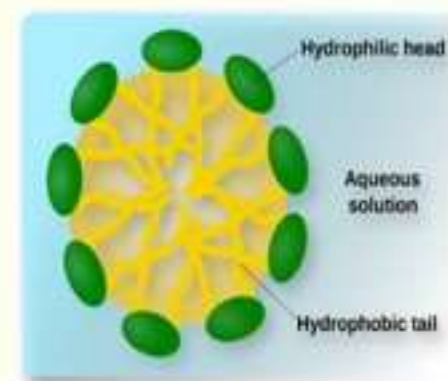
2 HYDROPHOBIC

Hydrophobic colloid
lack of attraction
between the two
phases. They often
coagulate or precipitate.
e.g. clay



3 ASSOCIATED

Association colloid-
association structure of
the dispersed phase
e.g.. Micelle of soap
molecules in water





IMPURE MATTER

COMMON METHODS OF SEPARATION OF MIXTURES

HANDPICKING



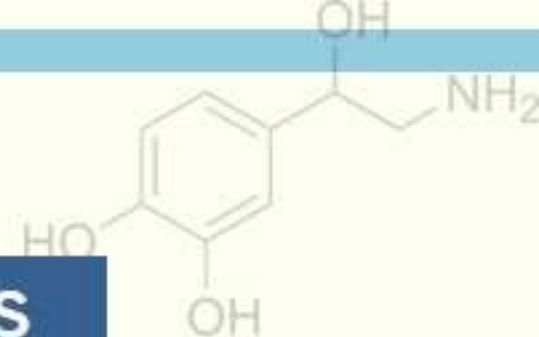
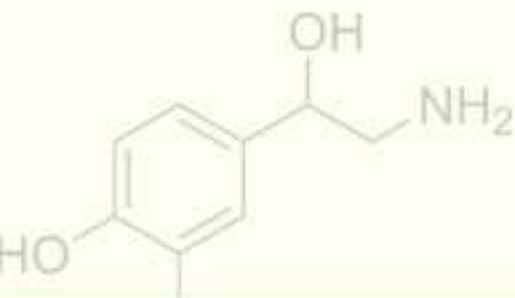
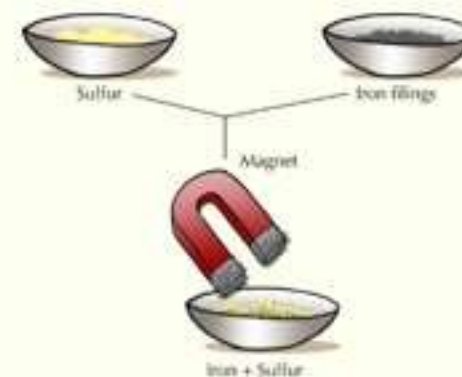
WINNOWING

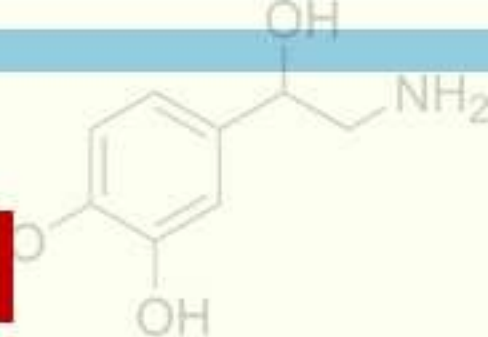


SIEVING



MAGNETIC SEPARATION





SEPARATION OF MIXTURES - COMMON METHODS

SEDIMENTATION AND DECANTATION

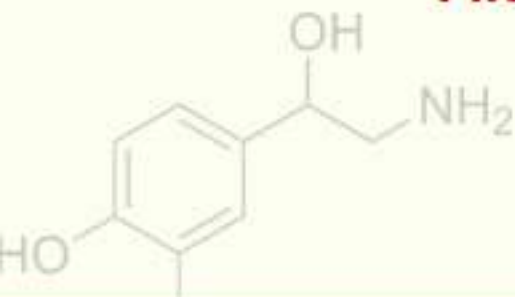


Filtering impure water

FROTH FLOATATION



Used in paper recycling and waste water treatments





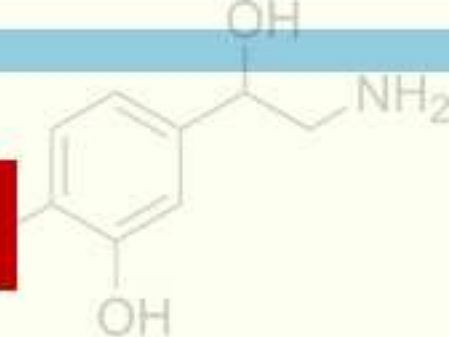
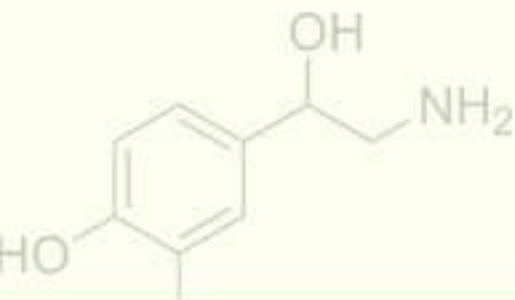
SEPARATING MIXTURES - WHAT IS THE USE?

1

MAGNETIC SEPARATION



Separates magnetic component from nonmagnetic one example to separate iron from Sulphur mixture

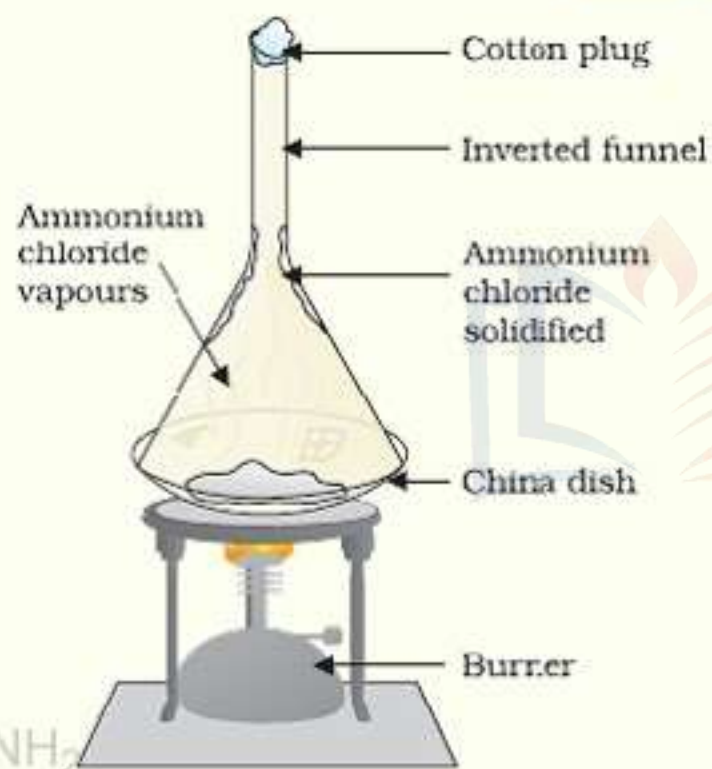




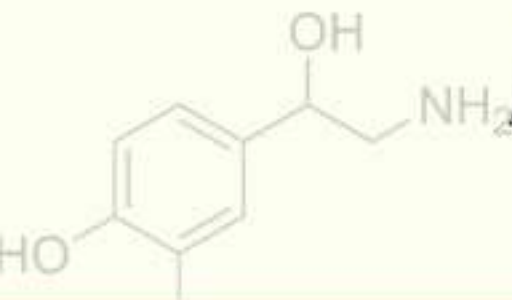
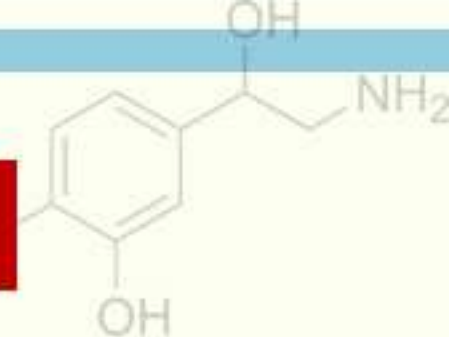
SEPARATING MIXTURES - WHAT IS THE USE?

2

SUBLIMATION



- Separate solute or volatile particle from non volatile solvent.
- A mixture of salt and ammonium chloride is separated in this way





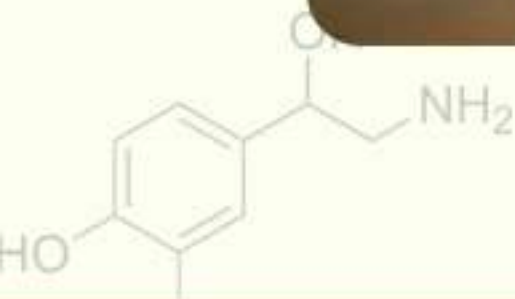
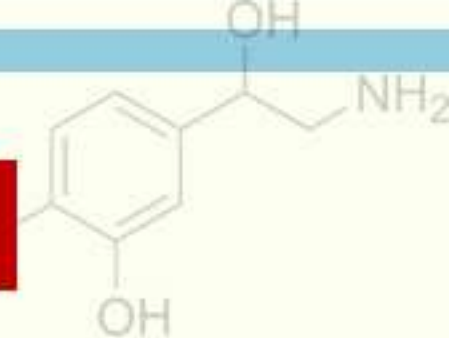
SEPARATING MIXTURES - WHAT IS THE USE?

3

EVAPORATION



- *Evaporation is a technique in which the solvent which is volatile becomes vapour and it leaves behind the non volatile solute.*
- *This way is largely used to separate sea salt from the sea water*

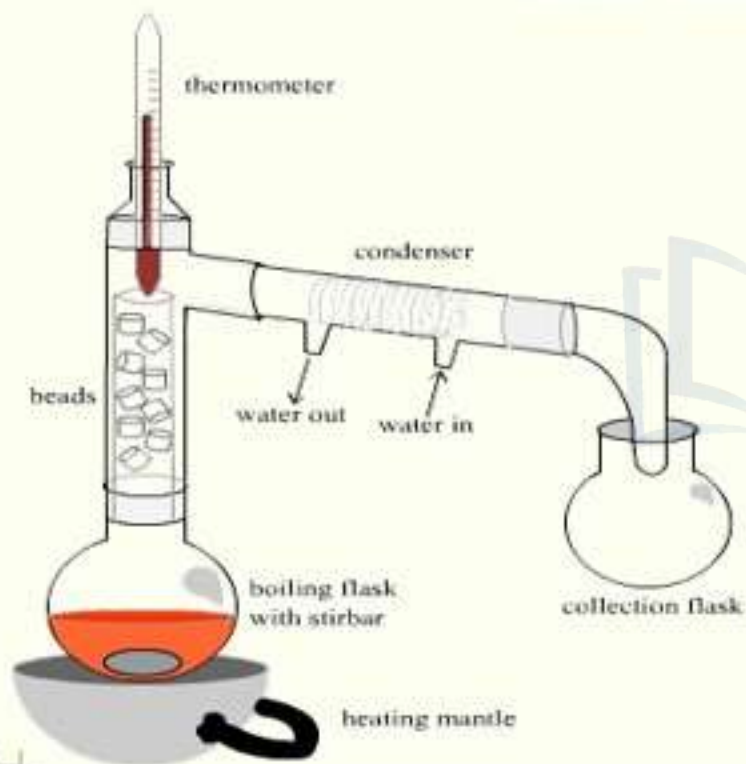




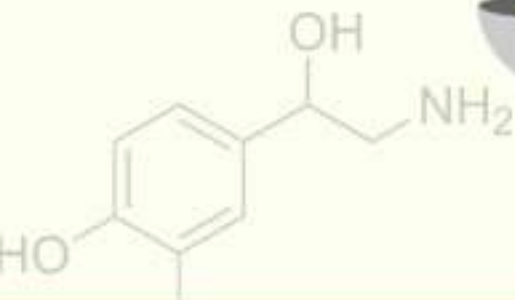
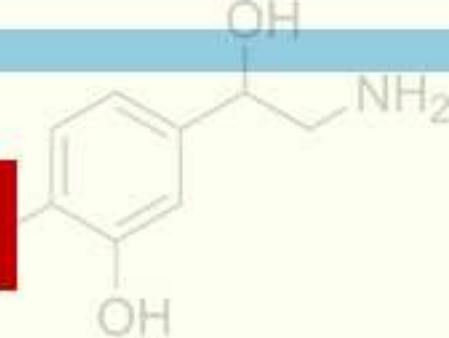
SEPARATING MIXTURES - WHAT IS THE USE?

4

DISTILLATION



- *To separate two miscible liquids that have sufficient difference in their boiling points.*
- *It is used to separate separation of petroleum components using fractional distillation*





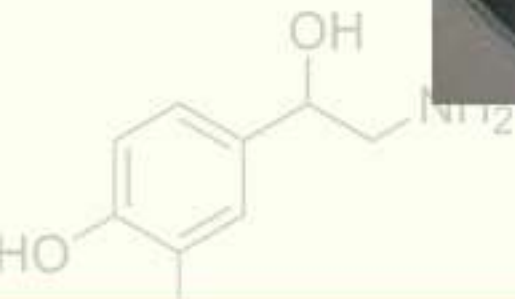
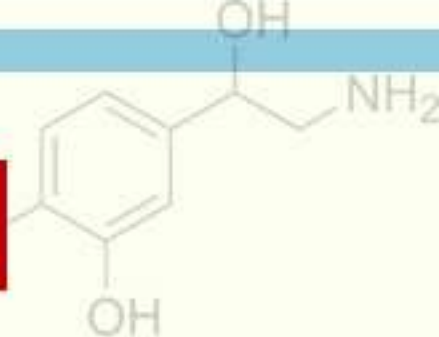
SEPARATING MIXTURES - WHAT IS THE USE?

5

CENTRIFUGATION



- The denser particle is separated from the lighter particle by means of heavy rotation.
- Centrifugation is used in diagnostic test for blood and urine
- Also is used in washing machines

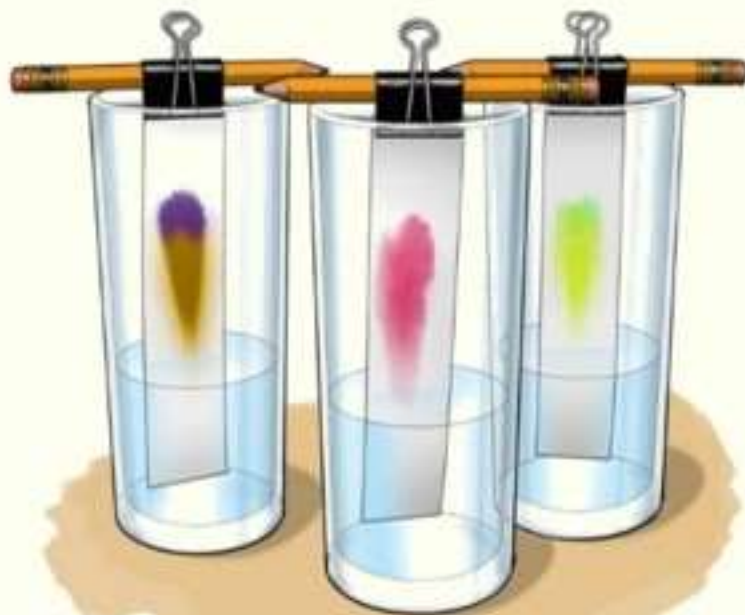




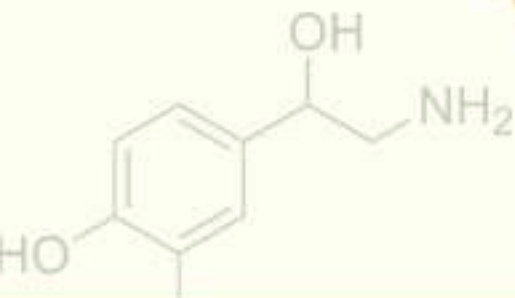
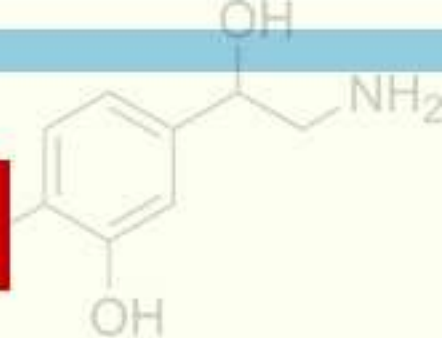
SEPARATING MIXTURES - WHAT IS THE USE?

6

CHROMATOGRAPHY



- *Chromatography is a technique which uses the solubility of different coloured pigments to separate coloured mixtures.*
- *Chromatography is used to separate from natural colours and drugs from blood*



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